Chart patterns Claude can identify — Trades tab, box 3

Full catalogue sent to Claude by `tr_pat_llm_prompt()` (`R/utils_patterns_llm.R`), grouped the way the course and the app's own code already group them (Step 1 and Step 2 are the app's own rule-based engine in `R/utils_patterns.R`; Steps 3–6 are the patterns added so Claude's scan is complete). Claude reads the same candles as the chart on screen, plus indicators computed deterministically in R (never invented by Claude): EMA 20/50/200, SMA20, RSI14, MACD + signal + histogram, Bollinger bands, Keltner channel, ADX14 with DI+/DI-, Stochastic %K/%D, ATR14.

Rules applied to every pattern, whichever step:

- **Multi-candle only.** A pattern on a single candle (hammer, doji, shooting star, marubozu, spinning top) is never asked for and is refused on the way back even if Claude names one.
- **Must reach the right-hand edge.** Only a pattern whose last candle sits within the final 10 candles is accepted; anything that finished earlier is dropped.
- Every pattern returns the level that **cancels** it (stop side) and its **measured target** (limit side), read off the structure itself (neckline, pole, band, moving average, measured move) rather than invented.
- `confidence` (0–100) and `status` (forming / confirmed / broken) are reported honestly; an empty list is a valid answer.
- A key not in this catalogue is refused; a key is reported once, at its best instance.

Source column: **App** = found today by the app's own deterministic rules (`utils_patterns.R`) as well as offered to Claude; **Claude** = only Claude's scan reports it.

Step 1 — Trend & market structure

Pattern	Bias	Source	Criteria Claude is told to apply
Uptrend (Dow Theory)	Bullish	App	Higher swing highs **and** higher swing lows across the window; "early" if not yet confirmed by a fresh higher high.
Downtrend (Dow Theory)	Bearish	App	Lower swing highs **and** lower swing lows across the window.
Support level	Bullish	App	A price shelf touched 3+ separate times (bounces, not one continuous move) without a clean close through it.
Resistance level	Bearish	App	Same test on the upside: 3+ separate touches that reject.
Old support now resistance	Bearish	App	A broken support level later capping price from above (role reversal).
Old resistance now support	Bullish	App	A broken resistance level later holding price from below.
Uptrend line	Bullish	App	3+ touches on rising swing lows, not steeper than ~0.6 ATR per candle (too steep = unsustainable, rejected).
Downtrend line	Bearish	App	3+ touches on falling swing highs, same slope test.
Broken uptrend line	Bearish	App	A valid uptrend line closed through from above, break still recent (last ~5 candles) and holding.
Broken downtrend line	Bullish	App	Mirror: a downtrend line closed through from below, recent and holding.
Broken countertrend line (uptrend continues)	Bullish	App	A line counter to the prevailing uptrend (drawn against the pullback) is broken upward: the main trend resumes.
Broken countertrend line (downtrend continues)	Bearish	App	Mirror, downtrend resumes after its countertrend line breaks down.
Bullish (ascending) channel	Bullish	App	Parallel rising trend line + a parallel upper line, both with 3+ touches.
Bearish (descending) channel	Bearish	App	Parallel falling trend line + parallel lower line, both with 3+ touches.
Bullish Engulfing	Bullish	Claude	2 candles: a down candle fully engulfed (open/close) by the next up candle's real body.
Bearish Engulfing	Bearish	App + Claude	Mirror: an up candle fully engulfed by the next down candle.
Piercing Pattern	Bullish	App + Claude	Down candle, then an up candle opening below its low and closing above its midpoint.
Dark Cloud Cover	Bearish	App + Claude	Up candle, then a down candle opening above its high and closing below its midpoint.
Bullish Harami	Bullish	Claude	A large down candle followed by a small up candle contained entirely within its body.
Bearish Harami	Bearish	Claude	A large up candle followed by a small down candle contained within its body.
Harami Cross	Trend	Claude	A harami whose second candle is a doji — indecision inside the prior range, direction set by what follows.

Pattern	Bias	Source	Criteria Claude is told to apply
Morning Star	Bullish	Claude	3 candles: a down candle, a small-bodied candle gapping/dipping lower, then an up candle closing well into the first candle's body.
Evening Star	Bearish	Claude	Mirror of Morning Star at a high.
Three White Soldiers	Bullish	Claude	3 consecutive up candles, each opening within the prior body and closing near its own high.
Three Black Crows	Bearish	Claude	3 consecutive down candles, each opening within the prior body and closing near its own low.
Tweezer Top	Bearish	Claude	2 candles sharing (near enough) the same high, first up then down.
Tweezer Bottom	Bullish	Claude	2 candles sharing the same low, first down then up.
Rising Three Methods	Bullish	Claude	A long up candle, 3 small down candles staying inside its range, then a new up candle closing above the first.
Falling Three Methods	Bearish	Claude	Mirror, continuation of a downtrend.
Three Inside Up	Bullish	Claude	A bullish harami followed by a third candle closing above the first candle's open.
Three Inside Down	Bearish	Claude	Mirror with a bearish harami.
Three Outside Up	Bullish	Claude	A bullish engulfing followed by a third candle closing even higher.
Three Outside Down	Bearish	Claude	Mirror with a bearish engulfing.
Bullish Kicker	Bullish	Claude	A down candle, then a gap up opening above the prior open, with no overlap.
Bearish Kicker	Bearish	Claude	Mirror, gap down with no overlap.
Stick Sandwich	Bullish	Claude	Down, up, down candles where the two down candles close at (near) the same price.
Matching Low	Bullish	Claude	Two down candles closing at (near) the same low, suggesting a floor.
Separating Lines (bullish)	Bullish	Claude	A down candle then an up candle sharing the same open, in an uptrend.
Separating Lines (bearish)	Bearish	Claude	Mirror in a downtrend.

Step 2 — Fibonacci & classic price patterns

Pattern	Bias	Source	Criteria
Fibonacci retracement	Trend	App	Pullback from the latest swing measured against 38.2/50/61.8% of the prior leg; the level price is reacting to is reported.
Fibonacci 100% expansion	Trend	App	The prior leg's length projected from the retracement low/high as a target.
Symmetrical triangle	Trend	App	Converging trend line (lower highs) and support line (higher lows), 5+ turning points, not overly steep.
Ascending triangle	Bullish	App	Flat resistance + rising support line, 5+ turning points.
Descending triangle	Bearish	App	Flat support + falling resistance line, 5+ turning points.
Rising wedge	Bearish	App	Both boundary lines rising, converging, upper line less steep (a weakening uptrend that usually resolves down).
Falling wedge	Bullish	App	Both boundary lines falling, converging, lower line less steep.
Bullish flag	Bullish	App	A steep pole (4+ ATR) up, then 5–15 candles drifting down in a tight parallel channel.
Bearish flag	Bearish	App	Mirror: steep pole down, then a shallow rising parallel channel.
Pennant	Trend	App	Steep pole (either direction), then a small symmetrical triangle instead of a parallel channel.
Double top	Bearish	App	Two peaks within ~1 ATR of each other, separated by a valley 3+ ATR deep, 4+ candles apart.
Double bottom	Bullish	App	Mirror at a low.
Head & Shoulders	Bearish	App	Three peaks, the middle (head) higher, the two shoulders roughly level and evenly spaced either side, neckline across the two troughs.
Inverse Head & Shoulders	Bullish	App	Mirror at a low, neckline across the two intervening highs.

Step 3 — Further price structures, gaps & retests

Pattern	Bias	Source	Criteria
Triple top	Bearish	Claude	Three peaks at (near enough) the same level, each separated by a real pullback.
Triple bottom	Bullish	Claude	Mirror at a low.
Rectangle / trading range	Trend	Claude	Price oscillating between a clear flat support and flat resistance, 3+ touches each.
Cup and handle	Bullish	Claude	A rounded U-shaped decline-and-recovery back to the prior high (the cup), then a short shallow pullback (the handle) before continuation.
Inverted cup and handle	Bearish	Claude	Mirror: a rounded rise-and-fall, then a small bounce before continuation down.
Rounding bottom (saucer)	Bullish	Claude	A slow, gradual U-shaped reversal with no sharp turning point.
Rounding top	Bearish	Claude	Mirror, a slow dome-shaped reversal.
Diamond top	Bearish	Claude	Volatility broadens then narrows at a high, forming a diamond outline, before a break down.
Diamond bottom	Bullish	Claude	Mirror at a low.
Broadening top (megaphone)	Bearish	Claude	Both a rising resistance line and a falling support line (diverging), higher highs and lower lows, at a market top.
Broadening bottom	Bullish	Claude	Mirror at a low.
Island reversal top	Bearish	Claude	A gap up, a cluster of candles isolated from the move either side by a gap, then a gap down.
Island reversal bottom	Bullish	Claude	Mirror at a low.
Measured move up	Bullish	Claude	An initial leg up, a consolidation, then a second leg projected to roughly the same length as the first.
Measured move down	Bearish	Claude	Mirror.
Round-number level	Trend	Claude	Price repeatedly reacting at a psychological round number (e.g. 11000, 11050) rather than a technically derived level.
Volatility squeeze / coil	Trend	Claude	Candle ranges and ATR visibly contracting over several candles, compressing into a tight coil before an expected breakout.
Failed breakout (bull trap)	Bearish	Claude	Price broke above resistance, failed to hold, and closed back below it within a few candles.
Failed breakdown (bear trap)	Bullish	Claude	Mirror below support.
Throwback to a broken level	Trend	Claude	After a breakout, price returns to retest the broken level from the new side before continuing.
Pullback to the breakout level	Trend	Claude	Same idea stated the other way round: a shallow retest of the level just broken.
Breakaway gap up	Bullish	Claude	A gap up that launches price away from a base/range with strong follow-through.

Pattern	Bias	Source	Criteria
Breakaway gap down	Bearish	Claude	Mirror.
Runaway (measuring) gap up	Bullish	Claude	A gap up in the middle of an already-established uptrend, often used to measure the move's remaining length.
Runaway (measuring) gap down	Bearish	Claude	Mirror.
Exhaustion gap up	Bearish	Claude	A gap up at the end of an extended uptrend, quickly reversing — a sign the move is running out.
Exhaustion gap down	Bullish	Claude	Mirror at the end of a downtrend.
Spike reversal top	Bearish	Claude	A sharp, narrow one- or two-candle spike to a new high, immediately reversed.
Spike reversal bottom	Bullish	Claude	Mirror at a low.

Step 4 — Harmonic & Elliott structures

Pattern	Bias	Source	Criteria
Bullish AB=CD	Bullish	Claude	Two roughly equal-length legs (AB and CD) in the same direction separated by a retracement (BC), completing at a harmonic support zone.
Bearish AB=CD	Bearish	Claude	Mirror at resistance.
Three drives up (exhaustion)	Bearish	Claude	Three successive higher highs of similar proportion, each a Fibonacci-like extension of the last — a classic exhaustion top.
Three drives down (exhaustion)	Bullish	Claude	Mirror at a low.
Bullish Wolfe wave	Bullish	Claude	Five swing points forming two roughly parallel channel lines that converge, with point 5 piercing the 1–3 trend line — target is the opposite channel line (the "sweet spot").
Bearish Wolfe wave	Bearish	Claude	Mirror.
Elliott impulse up (1–5)	Bullish	Claude	A five-wave advancing sequence (1-2-3-4-5) where wave 3 is not the shortest and wave 4 does not overlap wave 1.
Elliott impulse down (1–5)	Bearish	Claude	Mirror, five waves down.
Elliott ABC correction	Trend	Claude	A three-wave corrective sequence (A-B-C) against the larger trend, typically retracing part of the prior impulse.

Step 5 — Moving averages, momentum, volatility & trend strength

Indicators are computed in R (not estimated by Claude) and handed over as a table plus the values at the last candle.

Pattern	Bias	Source	Criteria
Golden cross (50 over 200)	Bullish	Claude	EMA50 crosses above EMA200.
Death cross (50 under 200)	Bearish	Claude	EMA50 crosses below EMA200.
Pullback to the moving average (uptrend)	Bullish	Claude	Price in an established uptrend dips to touch/retest EMA20 or EMA50 then holds.
Pullback to the moving average (downtrend)	Bearish	Claude	Mirror.
Moving averages stacked bullish	Bullish	Claude	EMA20 above EMA50 above EMA200, all rising — a clean trend alignment.
Moving averages stacked bearish	Bearish	Claude	Mirror, all falling.
Moving-average ribbon squeeze	Trend	Claude	EMA20/50/200 converging and flattening — a trend losing momentum or a range forming.
Price crossed above the moving average	Bullish	Claude	Close crosses from below to above EMA20 or EMA50.
Price crossed below the moving average	Bearish	Claude	Mirror.
Holy Grail long (ADX > 30, pullback to the 20 EMA)	Bullish	Claude	ADX14 above 30 (strong uptrend) with a shallow pullback to the EMA20, offering a continuation entry.
Holy Grail short	Bearish	Claude	Mirror in a strong downtrend.
MACD crossed above its signal	Bullish	Claude	MACD line crosses above the signal line.
MACD crossed below its signal	Bearish	Claude	Mirror.
MACD crossed above zero	Bullish	Claude	MACD line crosses the zero line upward (momentum turns positive).
MACD crossed below zero	Bearish	Claude	Mirror.
Bullish MACD divergence	Bullish	Claude	Price makes a lower low while MACD makes a higher low.
Bearish MACD divergence	Bearish	Claude	Price makes a higher high while MACD makes a lower high.
RSI overbought	Bearish	Claude	RSI14 above 70.
RSI oversold	Bullish	Claude	RSI14 below 30.
Bullish RSI divergence	Bullish	Claude	Price lower low, RSI higher low.
Bearish RSI divergence	Bearish	Claude	Price higher high, RSI lower high.
RSI bullish failure swing	Bullish	Claude	RSI dips below 30, rallies, pulls back without re-making the low, then breaks its own prior high.
RSI bearish failure swing	Bearish	Claude	Mirror above 70.
RSI crossed above 50	Bullish	Claude	RSI14 crosses the 50 midline upward (momentum turns bullish).

Pattern	Bias	Source	Criteria
RSI crossed below 50	Bearish	Claude	Mirror.
Stochastic crossed up from oversold	Bullish	Claude	%K crosses above %D while both are below ~20.
Stochastic crossed down from overbought	Bearish	Claude	%K crosses below %D while both are above ~80.
Bullish stochastic divergence	Bullish	Claude	Price lower low, stochastic higher low.
Bearish stochastic divergence	Bearish	Claude	Price higher high, stochastic lower high.
Momentum exhaustion	Trend	Claude	Price still pushing to new highs/lows while every momentum reading (RSI, MACD histogram, stochastic) is flattening or turning — a warning rather than a trade signal on its own.
Bollinger squeeze	Trend	Claude	Band width (bb_width) contracts to a multi-candle low — volatility compressed, a breakout likely.
Bollinger breakout up	Bullish	Claude	Close breaks and holds above the upper band after a squeeze.
Bollinger breakout down	Bearish	Claude	Mirror, below the lower band.
Walking the upper Bollinger band	Bullish	Claude	Several consecutive closes hugging/riding the upper band — strong trend, not yet overbought in the mean-reversion sense.
Walking the lower Bollinger band	Bearish	Claude	Mirror.
Snap back from the lower band	Bullish	Claude	Price tags the lower band then sharply reverts toward the middle band.
Snap back from the upper band	Bearish	Claude	Mirror.
Keltner channel squeeze	Trend	Claude	Bollinger bands pull inside the Keltner channel — the same squeeze idea on a volatility (ATR) band rather than a standard-deviation band.
Volatility expanding (ATR rising)	Trend	Claude	ATR14 trending up over several candles — bigger ranges, often around a breakout or news.
Volatility contracting (ATR falling)	Trend	Claude	Mirror — a calming market, often before a squeeze breakout.
ADX above 25: the trend is strong	Trend	Claude	ADX14 > 25, confirming a trend is actually in force (not just noise).
ADX below 20: no trend, range conditions	Trend	Claude	ADX14 < 20 — range/mean-reversion conditions, trend signals less reliable.
DI+ crossed above DI-	Bullish	Claude	The +DI line crosses above -DI (directional movement turns up), especially meaningful with ADX rising.
DI- crossed above DI+	Bearish	Claude	Mirror.

Step 6 — Course strategies

Pattern	Bias	Source	Criteria
Pivot point bounce (long)	Bullish	Claude	Price reacts upward off a classic pivot support (PP, S1, S2) worked out from the prior day's high/low/close, as taught on the course.
Pivot point bounce (short)	Bearish	Claude	Mirror off a pivot resistance (PP, R1, R2).
Overnight-range straddle, upper side	Bullish	Claude	A buy-stop breakout above the 22:00–06:00 London overnight high, in line with the course's overnight-range straddle strategy.
Overnight-range straddle, lower side	Bearish	Claude	Mirror, a sell-stop breakout below the overnight low.
Opening-range breakout up	Bullish	Claude	A stop entry above the high of the defined opening range once the session starts trading.
Opening-range breakout down	Bearish	Claude	Mirror, below the opening range's low.

Coverage: 141 patterns across 19 groups / the 6 steps above. The app's own rule-based engine (`utils_patterns.R`) independently finds 38 of these (all of Steps 1–2); Claude's scan (`utils_patterns_llm.R`, button "Ask Claude to find every pattern (multi-candle)" in Trades box 3) covers the full 141, including everything the app's rules already find (so Claude can confirm or sharpen them) plus the 103 patterns in Steps 3–6 that have no deterministic rule written for them yet.